

The next step up in complexity is the `Sample Model`. Like with `Switch`, you can create instances of it in the Django Admin. Unlike the switch, it isn't on all the time. Each call to `sample_is_active()` returns a Boolean based on a random value. When you create a `Sample`, you specify the percentage of time it is active. Note that subsequent calls to `sample_is_active()` return a new value, so if you want the same result in a view, you should store the return value and reuse it.

Finally, Waffle's most powerful mechanism is the flag. This allows you to roll out a feature for specific users, groups of users, users that pass a test, or a percentage of visitors. A `Flag` gets associated with a request and gets accessed within a view. Calling `flag_is_active()` checks whether a global override is on, whether the flag is in testing mode, whether the current user meets the flag's criteria, whether the current user has a cookie (they've met the criteria in the past), and whether there is a random chance the feature should be enabled. The `Flag Model` contains the following attributes:

- Flag name
- On for everyone
- Testing mode
- Percentage allowed
- On for superusers
- On for staff
- On for authenticated
- On for a specific language
- A list of groups
- A list of users

By combining these criteria, you get a lot of control over who sees your conditional feature.

In addition to the `Model` classes, Waffle provides view decorators, mixins, and template tags. There are several other feature flag libraries out there, but Waffle is one of the longest-standing ones and is rock solid.

13.5 Static sites

Way back in chapter 1, I explained the idea of a static site: one where each URL maps directly to a file on the web server's hard drive. These sites are quite performant, as they require no additional database queries and are easily cacheable. Sometimes, a dynamic site is overkill. Of course, if you're building a static site, you still want a consistent look and feel, and it would be great to define the nav bar once and re-use it. Static site generators are tools that allow you to compose websites. You define pages that include reusable components, like your nav bar, and then the generator mashes everything together and outputs a set of files for your site.

There are quite a few static site generators out there, and they each have their own syntax and quirks. You've just spent a whole lot of time learning Django's syntax and quirks, so why go somewhere else? Django Distill (<https://django-distill.com/>) is a